

Application form mySNF

Instrument **Scientific Exchanges**

Part 1: General Information

Basic data

Project Title	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)
Project title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)

Research Field	Life sciences
Main Discipline	30207 Ecology
University	Eidg. Anstalt für Wasserversorgung - EAWAG

Applicant(s)

Main Applicant	Carlos Melian
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Grant Application

Amount requested (CHF)	Total	9'000
Requested starting date	01.10.2018	
Duration (months)	3	

Attachments

CV and publication list	CV_CarlosMelian.pdf CV_CeciliaAndreazzi.pdf
Further documents for scientific events or research visits	Form_confirmation_self_declaration_Signed.pdf
Other annexes	Research_Visit_Plan.pdf

1. Responsible applicant

Last name	Melian
First name	Carlos
Function (title)	Group Leader
Academic degree	Dr. / PhD
Date of birth	23.08.1969
Gender	male
Civil Status	Married
Language	German
Nationality	Spain
Correspondence address of application	Address of workplace

Home address

Address supplement	
Street, No.	Seestrasse 79
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
Country	Switzerland

Address of institute

Name of Institution 1 (e.g. laboratory) *	Eawag
Continuation 1 (e.g. inst / dept.)	
Continuation 2 (e.g. University)	
Street, No.	Seestrasse 79
Address supplement 1 (e.g. building)	
Address supplement 2 (e.g. office)	
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
State, canton, etc.	
Country	Switzerland

Communication

Secretariat line	
Switchboard	
Direct line	+41 58 765 22 08
Fax office	
Home telephone number	
Cellphone	+41 78 722 88 73
Website	http://www.eawag.ch/about/personen/homepages/melianca/index
E-mail address	carlos.melian@eawag.ch

2. Applicants' employment

Information on employment and function at the anticipated starting date of the grant

Name	Melian, Carlos
Employment at the anticipated starting date of the grant	permanent contract
Level of employment %	100
Function in the context of this grant application	Head of (e.g. institute, department, center, clinic)
Professorship	Assistant professor with tenure track
Doctorate (PhD)?	Yes
Date of doctorate (PhD)	01.02.2005
PhD supervisor	
Country of doctorate	Spain
Remarks	
Further employments	

3. Basic data I

Original title	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)
Title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems (research visit)
Requested starting date	01.10.2018
Duration (months)	3
Research field	Life sciences
Main discipline	30207 Ecology
Sub-discipline(s)	30206 Environmental Research

4. Basic data II

Summary

Background and Rationale (similar to the scientific event) Joining theoretical and empirical evolutionary ecologists to develop and test theoretical models that incorporate the empirical patterns of reciprocal trait changes will be critical to improve our predictions on biodiversity dynamics and their responses to multiple disturbances in species- and interaction-rich ecological communities. The aim of the joined workshop-research visit is to integrate trait-species-space interactions data into new process-based coevolutionary metaecosystem models. Such integration might facilitate the exploration of patterns of coexistence of co-evolved species, their role in ecosystem functioning, and the risk of collapse of co-evolved ecosystems, which is of critical importance for three reasons. First, the extinction of species caused by human-generated disturbances occurs at an unprecedented rate, and yet most models predicting extinction rates often assume that species evolve independently of one another. Second, although eco-evolutionary approaches of species interactions are increasingly common, studies connecting co-evolving trait distributions within and between species at small scales to investigate the large-scale association between coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics are rare. Third, although many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes, there is a gap in understanding how ecological and coevolutionary dynamics in species-rich assemblages shape biodiversity and ecosystem function in a spatio-temporal context. Objectives and Methods Both the scientific event and the research visit will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits. We will join biologists skilled in theoretical ecology, evolutionary biology, and community ecology to discuss about the following two questions: Question 1: How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems? We aim to integrate the effects of ecological, coevolutionary, and migration dynamics in shaping biodiversity in meta-ecosystems. We will merge the meta-ecosystem perspective into existing models of coevolutionary interaction networks. We will implement scenarios to predict biodiversity distribution across space and time under different functional interactions (i.e., trait matching and exploitation barriers, assuming weak and strong reciprocal selection strengths. Metacommunity and meta-ecosystem theory approaches will complement the functional scenarios to understand how spatial processes governed by dispersal or niche width can alter the coevolutionary dynamics within communities by contributing to species trait variation and network structure and stability, and thus ecosystem functioning. Question 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems? Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the fitness consequences of the ecological interaction (i.e., competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions between species. Yet, how the relationship between species traits and coevolution at local scales and the arrangement of ecosystems at the landscape scale produce rapid biodiversity changes and coevolutionary hot spots is not well understood. We will run simulations under landscapes with different spatial arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local and the landscape level. Expected results and Impact
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	We see great potential in linking the research of the participants to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, the workshop aims to decipher the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements. This is a venue that can move forward the needed synthesis between the ecological and evolutionary processes that shape biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based coevolution models in species rich communities, the main aim of this joined scientific event and research visit, remain at a very incipient stage.
Keywords	Biodiversity dynamics
	Metacommunity theory
	Coevolution
	Theoretical ecology
	Biodiversity patterns
Language of correspondence	German
Financial administration	EAWAG Verwaltung zHv Frau Barbara Breu Rütli

5. University or research institution

University	Eidg. Anstalt für Wasserversorgung - EAWAG
Remarks	

6. Requested funding

Requested funding	Total (CHF)	Year 1
Total (CHF)	9'000	9'000

Research funds	Total (CHF)	Year 1
Travel & subsistence allowances	9'000	9'000
Total (CHF)	9'000	9'000
Total (%)	100%	100%

Details

Travel & subsistence allowances	Total (CHF)	Year 1
Andreazzi, Cecilia, Associate Researcher, Fundação Oswaldo Cruz, Rio de Janeiro, Brasil	9'000	9'000
Relation to research plan / Comments / Additions	Expenses per month (3 months) Housing 1750CHF Food 900CHF Local transportation 350CHF	
Total (CHF)	9'000	9'000
Total (%)	100%	100%

7. Timeframe of the scientific exchange

Starting date	15.10.2018
Place	Kastanienbaum, Switzerland
Duration (days)	75

8. Contents

General information

Aim and motivation	0
Organisation	0

Target audience

Expected number of people (total number of participants)	0
Remark	0

Active participants

Number of active participants from the academic field	0
Number of active participants from outside the academic field	0
Remark	0
Number of women	0
Number of PhD students	0
Number of students	0
Remark	0

9. General remarks on the project

Subject	Submitted in parallel with a scientific event
Communication	This research visit will be joined to a scientific event submitted in parallel.
Confidential	No